CLASS TEST

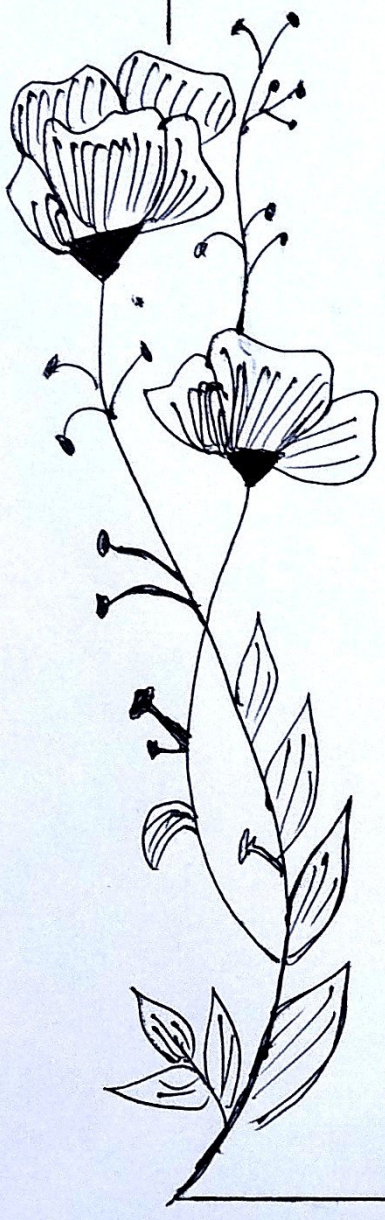
CO2 - AT3

COMPUTER VISION - ITA0501

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1) Grey-Level Transformation - Dark Museum photograph.

A dimly lit image can be enhanced using a log transformation or power-law (gamma) transformation. These transformations increase the intensity of dark pixels and make hidden details more visible.

power-law transformation:

$$S = cr^{\gamma}$$

where, r = input pixel intensity
S = output intensity
c = constant
γ = gamma value.

For $\gamma < 1$, dark regions are expanded and become brighter.

Processing Steps:

1. Read the dark image.
2. Convert the pixel values to a suitable range, usually 0-1
3. Apply the power-law transformation with $\gamma < 1$.
4. Rescale the resulting pixels to 0-255
5. Display the enhanced image.

Ex: If $\gamma = 0.5$, dark pixels are increased more strongly than bright pixels.

Result:

The shadow regions of the museum photograph become brighter, allowing details of exhibits to clearly seen.

27. Histogram Equalization - Fingerprint Image.

Histogram equalization is used when the pixel intensities are concentrated in a narrow range. It improves contrast and makes fingerprint ridges more visible.

The transformation is based on cumulative distribution function (CDF):

$$S_k = (L-1) \sum_{j=0}^{k} P_1(r_j)$$

Where,

- L = no. of intensity levels
- $P_1(r_j)$ = probability of intensity r_j
- S_k = new intensity value.

Processing steps:

1. Read the fingerprint image.
2. Calculate the histogram of pixel intensities
3. Normalize histogram to obtain probalities.
4. Calculate cumulative distribution function.
5. Map each old intensity to a new intensity using CDF
6. Obtain equalized image.

Result:

The narrow intensity distribution is spread over a wide range, increasing contrast between fingerprint ridges & valle

3) Salt-and-pepper Noise - Spatial Filtering

Salt-and-pepper noise produces isolated black and white pixels in an image. A median filter is suitable for removing the noise while preserving edges.

For example, for a 3×3 neighborhood, the pixels are arranged in increasing order and middle value (median) is selected.

processing steps:

1. Read the noisy image.
2. Select a median filter, such as 3×3.
3. Move the filter across the image.
4. Sort the neighbouring pixel values.
5. Replace centre pixel with median value.
6. Repeat for all pixels.

Why median instead of averaging?

An Averaging filter calculates mean and can blur important edges. A median filter removes isolated extreme values while preserving sharp boundaries better.

Result:

Salt-and-pepper noise is reduced while edges of objects remain relatively sharp.

4) Laplacian sharpening / Unsharp Masking Circuit Board

A Blurred PCB image can be sharpened using Laplacian-based sharpening to make copper tracks and their edges clearer.

The Laplacian operator detects rapid intensity changes.

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

A commonly used Laplacian mask is: $\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$.

processing steps:-

1. Read the Blurred PCB image
2. Apply Laplacian mask.
3. Detect high-frequency edge information.
4. Combine detected information with original image.
5. Obtain the sharpened image.

For example:-

$$g(x,y) = f(x,y) + \nabla^2 f(x,y)$$

depending on the chosen Laplacian convention.

Result:-

The edges of copper tracks become sharper and easier to inspect.

5) Frequency-Domain Low-pass Filtering - Drone Image

The drone image contains high-frequency grainy noise. A low-pass filter (LPF) can reduce this noise by allowing low frequencies to pass while attenuating high frequencies.

A Gaussian low-pass filter can be represented as:

$$H(u,v) = e^{-\frac{D^2(u,v)}{2D_0^2}}$$

Where D_0 is cutoff frequency.

Processing steps:-

1. Convert the image to frequency domain using Fourier Transform.
2. Shift the zero-frequency component to the centre.
3. Apply a low-pass filter.
4. Select an appropriate cutoff frequency.
5. perform inverse Fourier Transform.
6. Obtain the smoothed image.

Effect of cutoff frequency:

- Small cutoff: more smoothing and more noise removal, but details may be lost.
- Large cutoff: preserves more details but allows more noise.

6) Frequency - Domain High-pass / High-Boost Filtering - Retinal image

Thin blood vessels contain fine details and sharp intensity changes. A high-pass filter can enhance these high frequency components.

A high pass filter suppresses low frequencies and allows high frequencies to pass.

Processing Steps:

1. Read the retinal fundus image.
2. Apply Fourier Transform.
3. Shift the frequency spectrum.
4. Apply a higher-pass filter.
5. perform inverse Fourier Tranform.
6. Combine the enhanced details with original image if high-boost filtering is used.

High-boost filtering can be expressed as:

$$g(x,y) = Af(x,y) - f_{LP}(x,y)$$

Where $A \geq 1$

Result:

Fine vessel structures and their edges become more prominent against the background.

7) Point and line Detection - Bottle inspection.

Tiny air bubbles are isolated points, while hairline cracks can appear as thin lines. Therefore, point detection and line detection masks can be applied.

A point detection mask can be:

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

For line detection, different masks can detect horizontal, vertical and diagonal lines.

Example horizontal mask:

$$\begin{bmatrix} -1 & -1 & -1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{bmatrix}$$

Processing steps:

1. Apply point detection mask to identify isolated bubbles.
2. Apply horizontal, vertical and diagonal line masks.
3. Calculate the response with each mask.
4. Compare the response with a threshold.
5. Mark pixels whose response exceeds the threshold.

Threshold selection:

The Threshold should be high enough to reject normal surface variations but low enough to detect actual bubbles & cracks.

Result:

Air bubbles and hairline cracks are detected as distinct features.

8) Thresholding- Handwritten Answer Sheet.

The image contains dark handwriting on a white page, but the lighting is uneven. Therefore, adaptive thresholding is more suitable than simple global thresholding.

In global thresholding, one threshold T is applied to the entire image!

$$g(x,y) = \begin{cases} 1, & f(x,y) > T \\ 0, & f(x,y) \leq T \end{cases}$$

However, uneven illumination means different regions may require different thresholds.

Adaptive thresholding steps!

1. Read the handwritten image.
2. Select a local neighbourhood around each pixel.
3. Calculate a local threshold based on that neighborhood.
4. Compare the pixel with its local threshold.
5. Convert image into foreground and background.

Result!

Dark handwriting is seperated from the unevenly illuminated white background.

9) Region Growing - Satellite Lake

Region growing can be used to extract the lake because its pixels have nearly uniform intensity compared with the surrounding land.

Processing steps:

1. Select a seed point inside the lake.
2. Examine neighbouring pixels around the seed.
3. Compare their intensity with region's intensity.
4. Add pixels that satisfy the similarity criterion.
5. Continue checking neighbouring pixels.
6. Stop when neighbouring pixels no longer satisfy the criterion.

A similarity condition can be:

$$|f(x,y) - f_{seed}| < T$$

Where T is selected intensity tolerance.

Similarity criterion:

It should allow pixels with intensities similar to the lake while rejecting surrounding land.

Result:

The connected lake region is extracted from the Satellite image.

10) Edge Linking and Boundary Detection - Mechanical Gears

Noise and shadows can create broken edges in the detected gear image. Edge linking can connect these broken segments to obtain a complete gear boundary.

processing steps:

1. perform edge detection on the gear image.
2. Identify edge pixels.
3. Use local edge linking or Hough Transform to connect related edge points.
4. For a circular gear boundary to obtain the complete outline.

For a circle :-

$$(x-a)^2 + (y-b)^2 = r^2.$$

Where a, b represent the centre and r represents the radius of circle.

Result:

Broken edge segments caused by shadows and noise are connected, producing a more complete gear outline.